RO Skid - HIL Test Plan

What & How It Is Tested

EQSP32 Industrial IoT Controller

Document	Detail
Title	Reverse Osmosis Skid - Hardware-in-the-Loop Test Plan
Revision	v1.0
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Applies to	Firmware EQSP32_RO_Skid_Controller (HIL build)
Specification	RO Skid - Control & Operation Specification v0.3
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1. Purpose

This plan describes how the reverse-osmosis (RO) skid control firmware is validated against the RO Skid Control & Operation Specification using the EQSP32's built-in Hardware-in-the-Loop (HIL) facility. It documents the test environment, how each sensor is simulated, how the controller's decisions and outputs are observed, the time-compression used for the long process timers, and the complete test matrix with pass criteria.

2. Test Environment

Item	Value
Controller	EQSP32 (ESP32-S3), single unit, on-board I/O only
Firmware	EQSP32_RO_Skid_Controller.ino (HIL_TEST build)
Library	EQSP32 v1.3.2 (precompiled, esp32s3)
Toolchain	arduino-cli 1.5.0, esp32 core 2.0.17
Board (FQBN)	esp32:esp32:esp32s3:FlashSize=8M,PartitionScheme=default_8MB
Connection	USB serial, COM35 @ 115200 baud (8-N-1)
Test harness	run_hil_test.ps1 (PowerShell, System.IO.Ports.SerialPort)

3. HIL Method

The firmware calls eqsp32.beginHil() before eqsp32.begin(). In HIL mode the EQSP32 library runs the unmodified application logic, but the input/sensor reads (readPin) return values injected over USB serial instead of the physical front-end. Outputs set by the application are held in the normal output cache and can be read back. This lets the entire control program be exercised on the real silicon without any process plant connected.

Serial protocol used by the harness:

- Inject an input: -!EQ.<pin>=<raw> (e.g. -!EQ.1=1333 sets pressure-IN).
- Read an output: -?EQ.<pin> -> EQ.<pin>=<value> (output-cache duty 0 or 1000).
- Periodic frame: -!#<ms>/... re-applies a frame every <ms> milliseconds.
- Telemetry: the firmware emits a structured >>STATE line on every state/output change and once per second, carrying the controller state, the three actuator commands, the LED colour/flash code and all engineering values.

Full-vector periodic injection: the two flow meters use pulse-count (PCC) inputs, whose counts are consumed on each read. To keep them continuously fed, the harness re-sends the COMPLETE input vector as a single periodic frame (-!#300/...) whenever any input changes; the sticky analog/digital inputs are simply repeated harmlessly while the flow values are refreshed every 300 ms.

4. Signal Injection & Scaling

Each sensor is simulated by injecting its raw HIL value; the firmware applies the same scaling it would use for the real transmitter.

Sensor (pin)	Mode	Engineering scale	Raw HIL formula	Example
Pressure IN/OUT (1,2)	CIN	4-20 mA = 0-300 PSI	raw = mA x 100	175 PSI -> 1333
Depth clean/supply (3,4)	CIN	4-20 mA = 0-500 cm	raw = mA x 100	300 cm -> 1360
TDS IN (5)	AIN	0-5 V = 0-10000 mg/L	raw = mV	400 mg/L -> 200
TDS OUT (6)	AIN	0-5 V = 0-1000 mg/L	raw = mV	40 mg/L -> 200
Run/Stop (7)	DIN	1 = RUN, 0 = STOP	raw = 0/1	RUN -> 1
Motor temp (8)	TIN	10K NTC, deg C	raw = C x 10	25 C -> 250
Flow IN/OUT (9,10)	PCC	100 pulses = 1 L	L/min = pulses x 0.6	10.2 L/min -> 17
Outputs (11-16)	POUT	0 / 1000 duty	read-back	ON -> 1000

5. Observation & Verification

- Primary: the >>STATE telemetry line is parsed for controller state (st), pump/supply/flush commands (pump/sv/fv), LED colour and flash pattern (led/blink), active warning (warn) and latched fault with red blink code (fault/fcode).
- Secondary: actuator outputs are independently read back from the output cache via -?EQ.<pin> queries (pump=15, supply=16, flush=14) to confirm the physical command, not just the reported state.
- Each check waits for the expected condition (with a timeout) and records the actual telemetry snapshot, so both passes and failures capture real device data.

6. Time Compression (HIL build)

The control logic and all setpoints are identical to production. Only the long timers are scaled down so the full behaviour can be validated in under a minute; the production values are restored by building without HIL_TEST.

Timer	Production	HIL test
Start settling window	25 s	1.5 s
Minimum pump ON-time	10 s	1.5 s
Minimum pump OFF-time	10 s	1.5 s
Dry-run grace	10 s	1.5 s
Periodic flush interval	60 min	8 s
Flush duration	5 min	2 s
Permeate TDS trip delay	10 min	3 s
Analog fault debounce	5 s	0.6 s

7. Test Matrix

ID	Objective	Injected stimulus	Expected outcome
A1	Safe stop with healthy sensors	Run/Stop=STOP, all sensors nominal	STOP; pump+valves off; LED off
A2	Power-on / auto start (no operator start)	Run/Stop=RUN	STARTING then RUNNING; pump+supply valve on; LED green
A3	Output-cache cross-check	Query pump/supply/flush outputs	pump=1000, supply=1000, flush=0
B1	Clean tank full -> stop	Clean level = 405 cm	IDLE_FULL; pump off; LED blue solid
B2	Full hysteresis hold	Clean level = 385 cm	Stays stopped (no restart above 380)
B3	Restart on draw-down	Clean level = 375 cm	Auto restart to RUNNING
B4	Supply low -> pause	Supply level = 40 cm	PAUSE_SUPPLY; pump off; LED blue blink
B5	Supply hysteresis hold	Supply level = 55 cm	Stays paused (no resume below 60)
B6	Resume on refill	Supply level = 65 cm	Auto resume to RUNNING
C1	High feed pressure (early fouling)	Pressure IN = 210 PSI, normal permeate	WARNING; LED yellow slow
C2	Membrane cleaning required	Pressure IN = 210 PSI, low permeate	WARNING; LED yellow fast
C3	Low feed flow	Feed ~4.6 L/min	WARNING; LED yellow medium
C4	Recovery out of band	Recovery ~35% (>30)	WARNING; LED yellow medium
C5	Rising permeate TDS	TDS OUT = 180 mg/L	WARNING; LED yellow fast
D1	Motor over-temperature	Motor = 85 deg C	FAULT (code 1); pump off; LED red
D1c	Reset over-temperature	Run/Stop=STOP	Fault cleared; STOP
D2	Dry-run / no feed flow	Feed = 0 while running	FAULT (code 2); pump off
D2c	Reset dry-run	Run/Stop=STOP	Fault cleared; STOP
D3	Permeate TDS > 200 sustained	TDS OUT = 210 mg/L for the trip delay	FAULT (code 3); pump off
D3c	Reset TDS fault	Run/Stop=STOP	Fault cleared; STOP
D4	Over-pressure	Pressure IN = 260 PSI (>250)	FAULT (code 4); pump off
D4c	Reset over-pressure	Run/Stop=STOP	Fault cleared; STOP
D5	Broken-wire sensor	Pressure IN loop = 3.0 mA (<3.5)	FAULT (code 5); pump off
D5c	Reset sensor fault	Run/Stop=STOP	Fault cleared; STOP
E1	Emergency stop while running	Run/Stop=STOP during RUNNING	Immediate STOP; all outputs off; LED off
E2	E-stop clears a latched fault	Cause fault, then Run/Stop=STOP	STOP; fault cleared; LED off

F1	Minimum pump ON-time	Tank full immediately after start	Pump keeps running until min-on, then stops
F2	Minimum pump OFF-time	Demand run immediately after a stop	Restart deferred until min-off elapses
G1	Periodic membrane flush	Run continuously past flush interval	FLUSHING; pump on; flush valve open; LED yellow solid
G2	Flush valve output energised	Query outputs during flush	pump=1000, flush=1000
G3	Flush completes	After flush duration	RUNNING; flush valve closed; LED green

8. Pass Criteria

- A test passes only when the controller reaches the expected state, actuator commands and LED indication within the timeout, confirmed from live telemetry (and output-cache queries where noted).
- The build is accepted when 100% of the matrix passes. Any failure blocks acceptance and is investigated against the specification.

9. How To Run

Build and flash the HIL firmware, then execute the harness:

```
arduino-cli upload -p COM35 --fqbn
esp32:esp32:esp32s3:FlashSize=8M,PartitionScheme=default_8MB EQSP32_RO_Skid_Controller

powershell -ExecutionPolicy Bypass -File run_hil_test.ps1
```

The harness prints a PASS/FAIL line per check and writes TEST_REPORT.md with the full results.